

Few Puzzles in Physics

Presidency University Physics Society · 6 Lectures × 1 Hour

Course Description. This course traces the four important frameworks of modern physics — starting from 19th century to 20th century — through the specific puzzles that made each one necessary. Each lecture opens with a contradiction that the existing theory could not resolve, and closes with the conceptual structure forced upon physicists by that contradiction. The final lecture presents one of the deepest open problem of our time: the incompatibility of general relativity and quantum theory.

Lectures.

1. *Puzzles leading to EM and STR.* Faraday to Maxwell's work; the Michelson–Morley experiment; Einstein's two postulates and the Lorentz transformation.
2. *Puzzles leading to GR.* Incompatibility of Newtonian gravity with STR; the equivalence principle; why $\eta_{\mu\nu}$ must become a dynamical $g_{\mu\nu}(x)$; the Einstein field equations.
3. *Consequences of GR.* Perihelion precession, light deflection, gravitational redshift (consequence in Modified gravity theory), gravitational waves, black holes, and cosmology.
4. *Need for QFT formalism.* Blackbody radiation; negative probabilities in relativistic QM; pair production and variable particle number; causality violation and its resolution.
5. *Consequences of QFT.* Quantum Electrodynamics and $g - 2$; gauge symmetry (using group theory), observables (experiments in LHC).
6. *One of the current puzzles.* Problem with quantising Gravity; the black hole information paradox; string theory, and holography.

Learning Objectives. Students will identify the contradictions that motivated each major framework; trace the logical path from each puzzle to its resolution.

Learning Outcomes. Students will leave with a coherent conceptual map of theoretical physics beyond the standard syllabus, better equipped to read introductory graduate texts and to think about open problems.

Prerequisites. Classical mechanics, Electro-statics, Magneto-statics, introductory calculus, Einstein summation convention (check Wiki).

References.

- R. D'Inverno, J. Vickers, *Introducing Einstein's Relativity*, Oxford University Press, 2022.
 - A. K. Raychaudhuri, উচ্চতর গতিবিদ্যা, পশ্চিমবঙ্গ রাজ্য পুস্তক পর্ষৎ, Click here.
 - A. K. Raychaudhuri, *General Relativity, Astrophysics and Cosmology*, Springer. Borrow online.
 - R. P. Feynman et al., *The Feynman Lectures on Physics*, Vols. I–III. Pearson India, 2012.
 - A. Zee, *Quantum Field Theory in a Nutshell*, 2nd ed. Princeton University Press, 2010.
 - D. Tong, *Lectures on QFT and GR* (freely available at damtp.cam.ac.uk/user/tong).
 - S. M. Carroll, *Spacetime and Geometry*. Addison-Wesley, 2004.
 - A. Sen, *Short Lecture series on String theory*. Click here.
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EM: Electrodynamics, STR: Special Theory of Relativity, GR: General Relativity, QFT: Quantum Field Theory